Mirage (Desert / Military / Deception & Information Warfare)

Executive Summary

Mirage is a **military desert battlefield** built for **deception, misinformation, sensor manipulation, and autonomous counter-recon operations**. The terrain features heat-warped flats, dune fields, military outposts, radar stations, decoy emplacements, false infrastructure, buried cables, and canyon-edge observation posts. The map emphasizes Obsidian Protocol's autonomous deception logic, sensor-fusion under mirage distortion, comms-relay spoofing, and logistics under extreme heat and visibility anomalies. In AR (Nreal Light/Air), the map anchors at ~4–5 m viewing distance, covering a 10x10 m physical area representing ~1 km² of desert-military terrain. Key features include decoy zones, false sensor signatures, long-range mirage corridors, and information-warfare-driven engagements.

Design Goals & Gameplay

Deception-Driven Warfare

Mirage prioritizes false signals, decoy units, spoofed radar returns, holographic projections, and misinformation traps. Players manipulate enemy perception more than terrain.

Information Warfare & Sensor Manipulation

Heat shimmer, mirage distortion, dust haze, and thermal plumes degrade sensors. Specialized units generate false signatures, jam comms, or create phantom targets.

Autonomy & Counter-Recon Logic

Autonomous units choose routes based on sensor clarity, deception zones, false positives, and enemy spoofing attempts. Units adapt to misleading data and prioritize verification.

Desert Military Operations

Long-range fire, drone reconnaissance, armored pushes, and radar-guided strikes dominate. Outposts and radar stations serve as both real and decoy objectives.

Logistics & Resources

Fuel depots, comms bunkers, and desert roads serve as resupply nodes. Convoys must navigate exposed flats and avoid ambush-prone deception corridors.

Deployment & Progress

A fixed deployment budget (~10,000 points) forces players to choose between deception units, long-range firepower, or sensor-resistant armor. Blue deploys from the southwest dune-base staging area; Red deploys from the northeast radar-spire command.

Spatial Constraints (Nreal AR)

Device Capabilities

Nreal Light/Air: ~53° diagonal FOV, 1920×1080 per-eye resolution. Optimal viewing distance ~4–5 m.

Playable Area & Scaling

10×10 m physical area ≈ 1 km² desert-military zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

Inside-out tracking detects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000×1000 grid (1 unit ≈ 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ (Dune-Base Command) — (0,0), 150×150 m Armored vehicles, drone pads, deception-unit staging.
Red HQ (Radar-Spire Command) — (1000,1000), 150×150 m Elevated bunker with radar mast, AA emplacements, and sensor arrays.
Central Mirage Flats — (500,450), 300×300 m Heat shimmer, false signatures, long-range fire lanes, deception traps.
Comms Bunker Cluster — (250,800), 120×120 m Servers, generators, decoy transmitters, interior flanking routes.
Canyon Observation Network — (800,300), 200×200 m High-ground overwatch, sensor triangulation, counter-recon vantage points.
Deception Corridor — (0,450 → 400,900) Primary misinformation zone; false radar returns and phantom units.
Desert Road Route — (300,0 → 450,350) Safer but predictable; vulnerable to spoofed ambushes.
Radar Relay A — (650,150), +240 m Long-range sensor vantage point.
Radar Relay B — (150,650), +260 m Comms relay hub and deception-control node.
Neutral Forward Outpost — (500,950) Drone resupply zone.

Chokepoints & Sightlines

Deception Corridor

Major chokepoint; filled with false signals and spoofed targets.

Canyon Observation Network

Extreme long-range sightlines; ideal for counter-recon and sensor triangulation.

Mirage Flats

Long-range engagements distorted by heat shimmer and false signatures.

Desert Road Route

Predictable but safe; vulnerable to deception-driven ambushes.

Sightlines

Radar Relay A provides long-range sensor coverage (~400 m). Heat distortion and dust haze create fog-of-war pockets.

Deployment Zones

Blue Deployment: SW dune-base staging + deception units.
Red Deployment: NE radar-spire command + AA emplacements.
Neutral Forward Zone: Outpost at (500,950) for drone drops.

Assets & Environment

Terrain

Desert flats, dunes, canyons, roads, radar stations, comms bunkers.

Vegetation

Sparse shrubs, desert grass.

Buildings & Props

Military: AA guns, radar dishes, bunkers, decoy towers. Industrial: Generators, pipelines, comms arrays. Logistics: Trucks, crates, fuel drums, drones.

Effects

Heat shimmer, dust storms, mirage distortion, false holographic signatures.

Audio

Wind, distant machinery, radio chatter, drone hum.

LOD & Budget

Asset Cat	Count	Tri/Item	Total Tris
Desert Te	3	200k	600k
Industrial	4	80k	320k
Buildings/	15	50k	750k
Roads/Pa	4	30k	120k
Props	50	10k	500k
Total	—	—	**~2.29M**

Texture Budget

2048² for terrain/buildings, 1024² for props. DXT1/5 compression. Target ≤100 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal ARTemplate + Nreal SDK (XREAL). ARKit/ARCore enabled.

Rendering

Forward renderer, stationary lights, baked GI, aggressive culling.

Anchors

Single spatial anchor for entire map.

Interaction

Phone pointer for pan/zoom; HUD as screen-space widgets.

Multiplayer

Client-server architecture; offload AI/physics to server.

Implementation Plan & Testing

Milestones

1. **Prototype Layout — Month 1** Block-out mirage flats, canyon network, comms bunkers, AR anchor test.
2. **Core Systems — Month 2** NavMesh, deception autonomy behaviors, sensor-spoofing logic.
3. **AR Integration — Month 3** Anchors, plane detection, basic interactivity.
4. **Content Fill — Month 4–5** Final models, textures, lighting, audio.
5. **Gameplay & Polish — Month 6–7** Objectives, balance, VFX (heat shimmer, dust haze, holographic decoys).
6. **Testing & QA — Month 8** Multiplayer stress test, sensor validation.

Testing Plan

Autonomy

Verify deception-aware navigation, counter-recon behavior, hazard avoidance.

Comms/Sensors

Test relay destruction, fog-of-war updates, LOS checks under mirage distortion.

Performance

≥60 Hz AR render on target device.

QA Checklist

Navigation, deception logic, deployment budget enforcement, anchor drift, sensor accuracy.